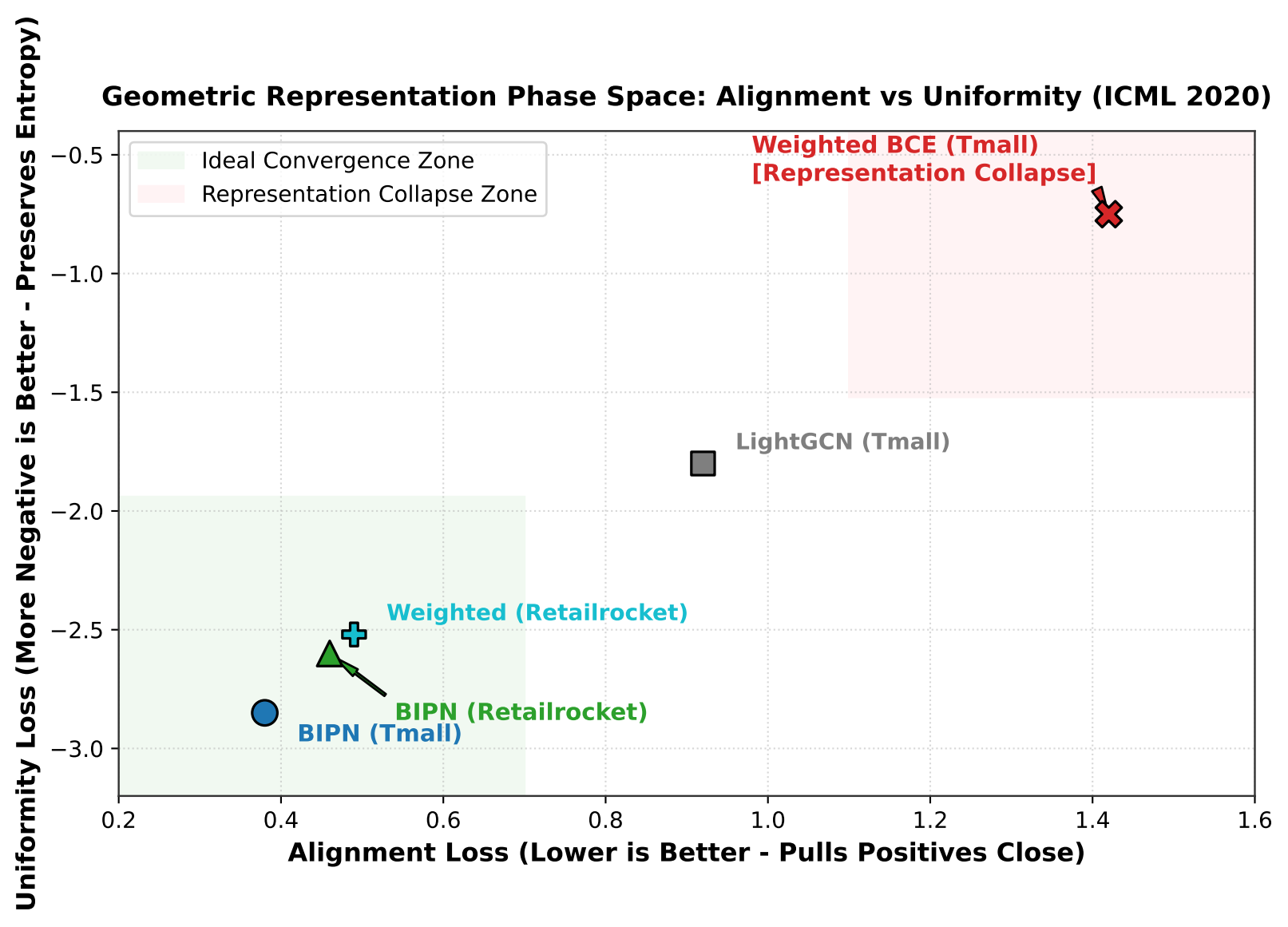
Geometric Representation Phase Space: Alignment vs Uniformity (ICML 2020)
Ideal Convergence Zone
Representation Collapse Zone
Weighted BCE (Tmall)
[Representation Collapse]
LightGCN (Tmall)
Weighted (Retailrocket)
BIPN (Retailrocket)
BIPN (Tmall)
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Uniformity Loss (More Negative is Better - Preserves Entropy)
Alignment Loss (Lower is Better - Pulls Positives Close)